



SPEED AND AGILITY · PRACTICAL GUIDE

+30 Speed, Acceleration and Change-of-Direction Workouts

A complete bank of routines to start faster, reach your top speed sooner and change direction without losing body control.

Acceleration

Top speed

Change of direction

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The science of acceleration and top speed

Accelerating and running at top speed are two different skills with different techniques. Most actions in a match use the first, not the second.



Acceleration (0-10 m)

Torso leaning forward, short powerful strides, foot strike under the center of gravity. This is the phase repeated most often in a match.



Transition (10-20 m)

The body gradually straightens up while the stride lengthens.



Top speed (beyond 20 m)

Almost vertical torso, maximum combined stride frequency and length. Rarely reached fully in a match.

<10 m

IS HOW LONG MOST REAL SPRINTS
LAST WITHIN A MATCH

2 techniques

DIFFERENT: YOU DON'T ACCELERATE
THE SAME WAY YOU RUN AT TOP SPEED

Hundreds

OF STARTS AND STOPS PER MATCH,
MORE THAN LONG SPRINTS



KEY TIP

If your goal is to perform better on the pitch, acceleration deserves more training volume than pure top speed: it's what you'll use most in a real match.



Why change of direction is a separate skill

Running fast in a straight line doesn't guarantee being fast on the pitch: football demands braking, reorienting the body and accelerating again, often in under a second.

- Braking efficiently requires eccentric strength: the ability to decelerate the body with control.
- The cutting angle (45°, 90°, 180°) completely changes the technique and the support needed.
- Reading the stimulus (opponent, ball, space) adds a cognitive component to pure agility.
- A poor braking and change-of-direction pattern is a known risk factor for knee and ankle injuries.



KEY TIP

Train braking as seriously as starting: a player who brakes well saves time and protects their joints on every change of direction.



How to use this routine bank

Speed is trained rested, not tired: the quality of each repetition matters far more than the quantity.

1. Always warm up with mobility, activation and running progressions before any routine in this ebook.
2. Respect the full rest periods indicated: fatigue ruins the technical quality of the sprint.
3. Prioritize the acceleration block if your training time is limited: it transfers best to the match.
4. Add 1 resisted or assisted speed session per week as a complementary stimulus.



WARNING

Never train top speed or intense changes of direction with high prior muscle fatigue: it is a known risk combination for muscle injuries, especially hamstrings.



BLOCK 1 OF 4

Acceleration: 0-10 meters

Eight routines to improve the phase repeated most often in a match: the explosive start over the first few meters.

8 routines · 15-20 minutes each

Full recovery between sets

Different starting positions



Acceleration: 0-10 meters

Improve the technique of the first steps: the phase repeated most often in a match.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Kneeling start	5-6	10 m	60-90 sec
	Start immediately after a vertical jump	5-6	10 m	60-90 sec
	Short hill acceleration	5-6	10 m	60-90 sec
	Seated start	5-6	10 m	60-90 sec



COACH'S TIP

The first 3-4 foot strikes are what most determine whether you win or lose the race: focus your attention there.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Acceleration: 0-10 meters

Train different starting positions to cover all real game contexts.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Kneeling start	5-6	10 m	60-90 sec
	Reactive start to a stimulus (whistle or clap)	5-6	10 m	60-90 sec
	Resisted start with a band held by a partner	5-6	10 m	60-90 sec
	Short hill acceleration	5-6	10 m	60-90 sec



COACH'S TIP

The forward torso lean at the start is key: a too-upright torso slows acceleration.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Acceleration: 0-10 meters

Build power in the start, with a leaning torso and powerful foot strikes.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Kneeling start	5-6	10 m	60-90 sec
	10 m acceleration from a standing start	5-6	10 m	60-90 sec
	Resisted start with a band held by a partner	5-6	10 m	60-90 sec
	Short hill acceleration	5-6	10 m	60-90 sec



COACH'S TIP

Practice starts from different positions: in a match you rarely start from sprint blocks.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Acceleration: 0-10 meters

Reduce reaction time to a stimulus (reactive start).

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Kneeling start	5-6	10 m	60-90 sec
	10 m acceleration from a standing start	5-6	10 m	60-90 sec
	Acceleration with a pronounced torso lean	5-6	10 m	60-90 sec
	Short hill acceleration	5-6	10 m	60-90 sec



COACH'S TIP

Take the full rest between sets: acceleration is trained fresh, not fatigued.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Acceleration: 0-10 meters

Transfer lower-body strength into a more explosive start.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Low-position start (three-point stance)	5-6	10 m	60-90 sec
	10 m acceleration from a standing start	5-6	10 m	60-90 sec
	Start immediately after a vertical jump	5-6	10 m	60-90 sec
	Short hill acceleration	5-6	10 m	60-90 sec



COACH'S TIP

The reactive start (to a stimulus) is the one that most resembles what really happens on the pitch.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Acceleration: 0-10 meters

Repeat the technical movement with quality, without accumulating fatigue between sets.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Low-position start (three-point stance)	5-6	10 m	60-90 sec
	Kneeling start	5-6	10 m	60-90 sec
	Acceleration with a pronounced torso lean	5-6	10 m	60-90 sec
	Resisted start with a band held by a partner	5-6	10 m	60-90 sec



COACH'S TIP

Don't chase top speed yet in this block: the focus is the explosiveness of the first meters.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Acceleration: 0-10 meters

Vary the starting point to simulate real game situations.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Kneeling start	5-6	10 m	60-90 sec
	Reactive start to a stimulus (whistle or clap)	5-6	10 m	60-90 sec
	10 m acceleration from a standing start	5-6	10 m	60-90 sec
	Acceleration with a pronounced torso lean	5-6	10 m	60-90 sec



COACH'S TIP

Add this block 1-2 times per week, away from match days.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Acceleration: 0-10 meters

Consolidate an efficient start with both legs as the initial push-off leg.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Low-position start (three-point stance)	5-6	10 m	60-90 sec
	Acceleration with a pronounced torso lean	5-6	10 m	60-90 sec
	Resisted start with a band held by a partner	5-6	10 m	60-90 sec
	Short hill acceleration	5-6	10 m	60-90 sec



COACH'S TIP

Film yourself from the side to check your torso angle at the start.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



BLOCK 2 OF 4

Top speed: 20-40 meters

Seven routines to raise your speed ceiling, the decisive head-to-head race against an opponent.

7 routines · 20-25 minutes each

2-3 minutes of recovery between sets

Ideally record times



Top speed: 20-40 meters

Raise your top-speed ceiling: the decisive head-to-head race against an opponent.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	30 m linear sprint	4-6	20-40 m	2-3 min
	Technical sprint focusing on stride frequency	4-6	20-40 m	2-3 min
	Assisted sprint on a gentle downhill	4-6	20-40 m	2-3 min
	Sprint repetitions with full recovery	4-6	20-40 m	2-3 min



COACH'S TIP

You need full recovery (2-3 minutes) between sets: top speed under fatigue loses the purpose of the training.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Top speed: 20-40 meters

Perfect running technique at maximum intensity: vertical torso, active arms.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	30 m linear sprint	4-6	20-40 m	2-3 min
	40 m linear sprint	4-6	20-40 m	2-3 min
	Sprint with timed splits	4-6	20-40 m	2-3 min
	Technical sprint focusing on stride length	4-6	20-40 m	2-3 min



COACH'S TIP

Technique tends to break down in the last meters of the sprint: pay special attention there.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Top speed: 20-40 meters

Work stride frequency and stride length separately before combining them.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Flying sprint (running start)	4-6	20-40 m	2-3 min
	Sprint with timed splits	4-6	20-40 m	2-3 min
	Technical sprint focusing on stride length	4-6	20-40 m	2-3 min
	Assisted sprint on a gentle downhill	4-6	20-40 m	2-3 min



COACH'S TIP

The flying sprint (a few meters of run-up before timing starts) lets you train real top speed, without the start phase.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Top speed: 20-40 meters

Record split times to measure your progress objectively.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	30 m linear sprint	4-6	20-40 m	2-3 min
	40 m linear sprint	4-6	20-40 m	2-3 min
	Technical sprint focusing on stride length	4-6	20-40 m	2-3 min
	Assisted sprint on a gentle downhill	4-6	20-40 m	2-3 min



COACH'S TIP

You don't need to do this block more than 1-2 times per week: it is highly demanding for the nervous system.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Top speed: 20-40 meters

Sustain technique even in the final stretch of the sprint, where it is most often lost.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	30 m linear sprint	4-6	20-40 m	2-3 min
	Technical sprint focusing on stride frequency	4-6	20-40 m	2-3 min
	Technical sprint focusing on stride length	4-6	20-40 m	2-3 min
	Sprint repetitions with full recovery	4-6	20-40 m	2-3 min



COACH'S TIP

A good technical sprint feels relaxed in the face and shoulders, even at top speed.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Top speed: 20-40 meters

Use maximum recovery between sets to maintain the quality of every repetition.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	30 m linear sprint	4-6	20-40 m	2-3 min
	Flying sprint (running start)	4-6	20-40 m	2-3 min
	Technical sprint focusing on stride length	4-6	20-40 m	2-3 min
	Sprint repetitions with full recovery	4-6	20-40 m	2-3 min



COACH'S TIP

If you feel a pull or muscle discomfort, stop the set immediately: don't force a sprint with compromised technique.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Top speed: 20-40 meters

Turn top speed into a reliable tool in long runs during the match.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	40 m linear sprint	4-6	20-40 m	2-3 min
	Flying sprint (running start)	4-6	20-40 m	2-3 min
	Technical sprint focusing on stride frequency	4-6	20-40 m	2-3 min
	Sprint repetitions with full recovery	4-6	20-40 m	2-3 min



COACH'S TIP

Record your times when you can: speed improvement shows over weeks, not in a single workout.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



BLOCK 3 OF 4

Change of direction and agility

Eight routines to brake, reorient the body and accelerate again: the skill that most sets an agile player apart.

8 routines · 15-20 minutes each

Bilateral work (both sides)

From fixed to reactive patterns



Change of direction and agility

Improve braking technique: the first step to changing direction efficiently.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Cone zigzag at top speed	5-6	10-20 sec per rep	45-60 sec
	T-drill agility circuit	5-6	10-20 sec per rep	45-60 sec
	180-degree change of direction	5-6	10-20 sec per rep	45-60 sec
	Agility with ball: dribbling and change of direction	5-6	10-20 sec per rep	45-60 sec



COACH'S TIP

Lower your center of gravity by bending hip and knee before each change of direction: it improves control and protects the joint.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Change of direction and agility

Train varied cutting angles (45°, 90°, 180°), like those repeated in a match.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	5-10-5 change of direction	5-6	10-20 sec per rep	45-60 sec
	Agility ladder with combined patterns	5-6	10-20 sec per rep	45-60 sec
	Stop-and-start drill	5-6	10-20 sec per rep	45-60 sec
	Cone circuit with multiple changes	5-6	10-20 sec per rep	45-60 sec



COACH'S TIP

Practice the patterns without a stimulus (planned) first, and only then add the reactive component.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Change of direction and agility

Add the reactive component: decide the direction in the moment, not in advance.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Agility ladder with combined patterns	5-6	10-20 sec per rep	45-60 sec
	Change of direction reactive to a visual stimulus	5-6	10-20 sec per rep	45-60 sec
	Cone circuit with multiple changes	5-6	10-20 sec per rep	45-60 sec
	Agility with ball: dribbling and change of direction	5-6	10-20 sec per rep	45-60 sec



COACH'S TIP

The 5-10-5 drill is one of the most complete: it combines acceleration, braking and change of direction in both directions.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Change of direction and agility

Reduce the injury risk associated with a poor braking and turning pattern.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	5-10-5 change of direction	5-6	10-20 sec per rep	45-60 sec
	Cone zigzag at top speed	5-6	10-20 sec per rep	45-60 sec
	Agility ladder with combined patterns	5-6	10-20 sec per rep	45-60 sec
	180-degree change of direction	5-6	10-20 sec per rep	45-60 sec

**COACH'S TIP**

Always work both sides equally: it's common to have one turning direction stronger than the other.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Change of direction and agility

Combine pure agility with ball dribbling to bring it closer to the real game.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Change of direction reactive to a visual stimulus	5-6	10-20 sec per rep	45-60 sec
	Stop-and-start drill	5-6	10-20 sec per rep	45-60 sec
	Cone circuit with multiple changes	5-6	10-20 sec per rep	45-60 sec
	Agility with ball: dribbling and change of direction	5-6	10-20 sec per rep	45-60 sec

**COACH'S TIP**

Quality of foot strike (full foot, aligned knee) matters more than pure speed in this block.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Change of direction and agility

Increase the speed at which the body reorients after each stop.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	5-10-5 change of direction	5-6	10-20 sec per rep	45-60 sec
	Change of direction reactive to a visual stimulus	5-6	10-20 sec per rep	45-60 sec
	180-degree change of direction	5-6	10-20 sec per rep	45-60 sec
	Cone circuit with multiple changes	5-6	10-20 sec per rep	45-60 sec



COACH'S TIP

Add the ball once you've mastered the pattern without it: dribbling completely changes the demand.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Change of direction and agility

Progress from fixed (planned) patterns to reactive (stimulus-driven) patterns.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	5-10-5 change of direction	5-6	10-20 sec per rep	45-60 sec
	Agility ladder with combined patterns	5-6	10-20 sec per rep	45-60 sec
	T-drill agility circuit	5-6	10-20 sec per rep	45-60 sec
	180-degree change of direction	5-6	10-20 sec per rep	45-60 sec

**COACH'S TIP**

This block is demanding on ankles and knees: don't do it in footwear unsuitable for the surface.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Change of direction and agility

Work agility to both sides equally.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Cone zigzag at top speed	5-6	10-20 sec per rep	45-60 sec
	Agility ladder with combined patterns	5-6	10-20 sec per rep	45-60 sec
	T-drill agility circuit	5-6	10-20 sec per rep	45-60 sec
	Cone circuit with multiple changes	5-6	10-20 sec per rep	45-60 sec



COACH'S TIP

Rest enough between repetitions to sustain maximum execution speed.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



BLOCK 4 OF 4

Resisted and assisted speed

Seven complementary routines with sled, bands, hills and the contrast method to add specific strength to your speed.

7 routines · 15-20 minutes each

Complementary, not a substitute

1 time per week recommended



Resisted and assisted speed

Add a strength stimulus specific to the running movement, with extra load or resistance.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Sprint with a resistance band held by a partner	4-5	10-20 m	90 sec - 2 min
	Sprint with a resistance parachute	4-5	10-20 m	90 sec - 2 min
	Assisted sprint on a very gentle downhill	4-5	10-20 m	90 sec - 2 min
	Multi-jumps followed by an immediate sprint (contrast)	4-5	10-20 m	90 sec - 2 min



COACH'S TIP

The sled or band load should be moderate: if it changes your running technique too much, it is excessive.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Resisted and assisted speed

Improve the power of each foot strike by working against moderate resistance.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Sprint with a resistance band held by a partner	4-5	10-20 m	90 sec - 2 min
	Steep hill sprint	4-5	10-20 m	90 sec - 2 min
	Sprint with weighted vest (light load)	4-5	10-20 m	90 sec - 2 min
	Multi-jumps followed by an immediate sprint (contrast)	4-5	10-20 m	90 sec - 2 min



COACH'S TIP

Assisted sprinting (gentle downhill) lets you experience a higher speed than you can reach on flat ground: useful to "feel" top speed.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Resisted and assisted speed

Use the contrast method (strength + speed) to boost acceleration.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Steep hill sprint	4-5	10-20 m	90 sec - 2 min
	Assisted sprint on a very gentle downhill	4-5	10-20 m	90 sec - 2 min
	Sprint with weighted vest (light load)	4-5	10-20 m	90 sec - 2 min
	Multi-jumps followed by an immediate sprint (contrast)	4-5	10-20 m	90 sec - 2 min



COACH'S TIP

A very steep hill sprint trains more strength than speed: don't confuse it with pure speed work.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Resisted and assisted speed

Vary the speed stimulus to avoid stagnation from your usual training.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Sled sprint (sled push or pull, light load)	4-5	10-20 m	90 sec - 2 min
	Sprint with a resistance parachute	4-5	10-20 m	90 sec - 2 min
	Assisted sprint on a very gentle downhill	4-5	10-20 m	90 sec - 2 min
	Multi-jumps followed by an immediate sprint (contrast)	4-5	10-20 m	90 sec - 2 min



COACH'S TIP

The contrast method (jump + immediate sprint) is advanced: master each component separately first.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Resisted and assisted speed

Reinforce running technique under a greater-than-usual strength demand.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Sled sprint (sled push or pull, light load)	4-5	10-20 m	90 sec - 2 min
	Sprint with a resistance parachute	4-5	10-20 m	90 sec - 2 min
	Steep hill sprint	4-5	10-20 m	90 sec - 2 min
	Sprint with weighted vest (light load)	4-5	10-20 m	90 sec - 2 min



COACH'S TIP

This block complements the previous ones: don't use it as your only speed stimulus of the week.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Resisted and assisted speed

Complement, not replace, the free-speed work of the previous blocks.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Sled sprint (sled push or pull, light load)	4-5	10-20 m	90 sec - 2 min
	Sprint with a resistance band held by a partner	4-5	10-20 m	90 sec - 2 min
	Steep hill sprint	4-5	10-20 m	90 sec - 2 min
	Sprint with weighted vest (light load)	4-5	10-20 m	90 sec - 2 min



COACH'S TIP

Progress the resistance (weight, tension) only when your running technique stays solid.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)



Resisted and assisted speed

Progress the load (sled weight, band tension) gradually.

	EXERCISE	SETS	DISTANCE / DURATION	REST
	Sprint with a resistance band held by a partner	4-5	10-20 m	90 sec - 2 min
	Sprint with a resistance parachute	4-5	10-20 m	90 sec - 2 min
	Steep hill sprint	4-5	10-20 m	90 sec - 2 min
	Multi-jumps followed by an immediate sprint (contrast)	4-5	10-20 m	90 sec - 2 min



COACH'S TIP

1 session of this block per week is usually enough as a complementary stimulus.

LOG (DATE · LOAD/INTENSITY · RPE 1-10 · NOTES)

TO CLOSE

Faster, in every meter that matters

A footballer's real speed isn't measured over 100 flat meters: it's measured by how long it takes you to win the first step against your opponent, and how well you control your body when you have to brake and start again. These 30 routines train both.



Prioritize acceleration: it is the phase repeated most often in a real match.



Always train top speed with full recovery between sets.



Don't neglect change of direction: braking well can also be trained.



Add resisted or assisted speed as a complementary stimulus, not the main one.



Never train speed with high muscle fatigue: it is a known injury risk factor.



This material is for educational and general training purposes. Maximum-intensity speed work carries a risk of muscle injury: make sure to warm up properly and consult a professional about any discomfort.